



PROFILE

Scientific leader with extensive experience connecting engineering, physics, materials science and applied research. Experienced in leading interdisciplinary collaborations, managing research projects and funding, developing talent, and translating complex scientific challenges into effective shared solutions. Passionate about building research infrastructures that enable outstanding science through excellent service, training and collaboration.

ASSETS

- CHF 737k competitive research funding managed.
- 15+ years interdisciplinary research.
- IP expertise.
- Extensive teaching & 30+ student supervisions.
- Scientific communication in four languages.

CONTACT DETAILS

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PERSONAL INFORMATION

Year of birth: **1985**

Citizenship: **The Netherlands**

Residency: **Switzerland** (C-permit)

Civil status: **Married**

Languages:

English (C2), **French** (C2), **German** (C2/C1), **Dutch** (native)

EXPERIENCE

- **PATENT EXPERT & EXAMINER** **2024–pres.**
Swiss Federal Institute of Intellectual Property (Switzerland)
 - Managed a large portfolio of technically diverse patent dossiers under strict legal deadlines.
 - Communicating with inventors and patent attorneys.
 - Analysing technologies across many disciplines.
 - Understanding innovation ecosystem.
 - Prioritisation under deadlines.
 - Member of communication task force.
- **SENIOR RESEARCHER PHYSICS** **2019–2024**
EPFL (Switzerland) – Funding: FNSF Ambizione (Switzerland)
 - Acquired and managed CHF 567k SNSF Ambizione project.
 - Established and coordinated interdisciplinary collaborations.
 - Supervised bachelor, master and PhD researchers.
 - Managed project planning, budgets and deliverables.
 - Conducted internationally recognised research.
 - Subjects: tribology, fracture, earthquakes, glasses, complex systems.
 - Proposal of a novel, interdisciplinary, theory of friction.
 - Delivery of top journal publications and conference presentations.
 - Design, implementation, execution high-performance computations.
 - Reviews for top journals.
 - Data-champion.
- **POSTDOCTORAL RESEARCHER PHYSICS** **2016–2019**
EPFL (Switzerland) – Funding: NWO Rubicon (The Netherlands)
 - Acquisition and management of funding (160 k€).
 - Delivery of top journal publications and conference presentations.
 - Co-supervision bachelor students.
 - Teaching assistant statistical physics (physics).
 - Teaching assistant continuum mechanics (civil engineering).
- **POSTDOCTORAL RESEARCHER VALORIZATION** **2016**
M2i & Tata Steel (The Netherlands)
 - Acquisition and management of funding (10 k€).
 - Developed industrial tool for scanning electron microscopy.
- **RESEARCH INTERNSHIP** **2010**
Harvard University (United States)
 - Design and analysis of a 3D meta-material.
 - Development of numerical tools.

EDUCATION

- **PHD MECHANICAL ENGINEERING (CUM LAUDE)** **2012–2016**
Eindhoven University of Technology (The Netherlands)
 - Part of, experimental-theoretical, Public-Private-Partnership (Eindhoven University, Material Innovation Institute M2i, Tata Steel).
 - Experimental-theoretical research on fracture of advanced steels in close collaboration with electron microscopy specialists.
 - Topic: mechanics & fracture of high-grade steel.
 - **Martinus van Marum award (12.5 k€) for: Best PhD thesis in five years in the engineering sciences**, and various other awards.
 - Delivery of journal publications and conference presentations.
 - Co-supervision of bachelor and master students.
 - Teaching finite element method, programming, and design-based learning.
- **MSc/BSc MECHANICAL ENG. (GREAT APPRECIATION)** **2004–2012**
Eindhoven University of Technology (The Netherlands)
 - Specialization: mechanics of materials.
 - Elective courses mathematics (numerical solvers).
 - Minor: entrepreneurship.
 - Winner entrepreneurship award (1 k€).

PROFESSIONAL STRENGTHS

- **ANALYTICAL THINKING:**
 - Combining theoretical physics & mechanics.
 - Designing and performing high-performance computing based on collaborative open-source tools.
- **CRITICAL THINKING:**
 - [Proposing a novel theory of friction](#) (despite it being an old and vast subject).
 - Performing patent assessments (and considering solutions in case of problems).
- **CREATIVITY:**
 - Combining knowledge from various disciplines.
 - Proposing novel simple models by abstraction.
- **ADAPTABILITY:**
 - Integrating in Switzerland.
 - Developing proficiency in various scientific fields.
 - Transitioning between various professions (academia and public federal administration).

PROJECT MANAGEMENT EXPERIENCE

- **UNITING:**
 - Initiating and fostering various interdisciplinary collaborations.
- **PRIORITISING & ACCOUNTABILITY:**
 - Balancing ad-hoc and permanent solutions.
 - Anticipating deadlines, co-workers and other third parties.
- **ACHIEVING RESULTS, PERSISTENCE & RESILIENCE:**
 - Accomplishing goals [Ambizione](#) grant.
 - Publishing in PNAS, Physical Review Letters, and other top journals (900+ citations on Google Scholar).
 - (Non-professional) [Becoming Swiss rowing champion](#).
- **BUDGETING:**
 - Managing research project budgets.
 - (Volunteering) Managing sport club budget.

LEADERSHIP EXPERIENCE

- **INSPIRING:**
 - Proposing and directing challenging student projects.
 - Finishing [world's longest rowing race on closed water](#), twice.
 - Performing open science & [open-source coding](#).
- **MOTIVATING, GUIDING, STRATEGISING, DECISION-MAKING:**
 - (Co)-Supervising a PhD student, and [32 bachelor and master projects](#).
 - (Volunteering) Leading mountaineering & rowing outings (both high-risk sports).
 - (Student job) Being a high school tutor (mathematics and physics).
 - Initiating a social awareness game at the beginning of the COVID-pandemic: [TV-interview RTS](#).
- **COMMUNICATING:**
 - Recognised for creating collaborative, open and supportive research environments.
 - Authoring scientific publications and giving conference presentations and seminars.
 - Performing patent-assessments and novelty analyses.
 - Providing an interview with the [FNSF Horizons](#) magazine.
- **DELEGATING, TEAM DEVELOPMENT:**
 - Running various scientific collaborations.
 - (Student job) Being a shift leader of a catering team in the football stadium of PSV.
 - Taking a motivating role in a team (professionally and in sports).
- **INCLUDING:**
 - (Volunteering) Being a buddy of a student on the autistic spectrum.
 - (Volunteering) Training of diverse rowers.
 - (Volunteering) Training of field-hockey keepers.